

8bi	$(-5.45 + 5.00) \times 10^{-19} = -0.45 \times 10^{-19} \text{ J}$ Hence the highest energy level which the atom can be excited to is A_4 .	A1
8bii	Highest energy photon emitted from A_4 to A_1 transition: $\frac{hc}{\lambda} = E_4 - E_1$ $\frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{\lambda} = [(-0.78) - (-5.45)] \times 10^{-19}$ $\lambda = 4.26 \times 10^{-7} \text{ m}$	C1 A1
8biii	Visible light (accept blue / violet light) Must tally with answer in (b)(bii)	A1
8ci1	The <u>minimum</u> energy required to eject an <u>electron</u> from the <u>metal surface</u> .	A1
8ci2	KE is maximum when electrons at the surface are ejected. KE of emitted electrons can be lower than the maximum energy because energy is required to bring an electron deeper in the metal to the surface / electron loses energy when it makes its way to the surface (upon collisions with other electrons and metal lattice), hence even though the energy absorbed by each electron from a photon is the same, the electrons can be emitted from the surface of the metals with a range of kinetic energies.	B1 B1
8cii1	$\frac{hc}{\lambda} = \phi + E_{\max}$ $\frac{1}{\lambda} = \frac{\phi}{hc} + \frac{E_{\max}}{hc}$ When $\frac{1}{\lambda} = 0$, ie. at x-intercept, $\phi = -E_{\max}$ Extend line to intersect on x-axis $\phi = 4.00 \times 10^{-19} \text{ J (allow } \pm 0.2 \times 10^{-19} \text{ J)}$	M1 A1
8cii2	$\frac{hc}{\lambda} = \phi + E_{\max}$ $\frac{1}{\lambda} = \frac{\phi}{hc} + \frac{E_{\max}}{hc}$ Gradient of line = $\frac{1}{hc}$ $\text{Gradient} = \frac{(4.00 - 2.50) \times 10^6}{(4.10 - 1.10) \times 10^{-19}} = 5.00 \times 10^{24}$ $h = \frac{1}{(5.00 \times 10^{24})(3.0 \times 10^8)}$ $= 6.67 \times 10^{-34} \text{ J s}$	M1 M1 A1